

Mohamed Dhia BEN HMIDA

Senior Technical Consultant | Tech Lead

Tunis, Tunisia

Email: m.dhia.b.h@gmail.com

Phone: +(216) 27 225 432 | +(216) 95 410 551

LinkedIn: linkedin.com/in/mohamed-dhia-ben-hmida-11b018135

Portfolio: <https://mdbh.netlify.app>

PROFESSIONAL SUMMARY

Senior technical consultant and tech lead with 8 years of experience designing mission-critical systems for banks. Tech lead at ITSS Global since March 2026, accountable end to end for regulatory reporting and enterprise document management deliveries in North and Central Africa: functional framing with business stakeholders, data architecture from core banking extraction through a Bronze / Silver / Gold lakehouse to report restitution, BPM-driven document workflows, and Nostro account reconciliation. Core stack: Java 21, Spring Boot, Oracle SQL and PL/SQL, PostgreSQL, Temenos T24 / Transact, React.

SKILLS

Programming: Python, JavaScript, TypeScript, Java, PHP, PL/SQL, VBA

Web development: React, FastAPI, Express (Node.js), Spring Boot, Ant Design, HTML, Bootstrap, Material UI

State management: Redux Toolkit

Databases: PostgreSQL, SQL Server, Oracle SQL, MySQL, MongoDB, JPA/Hibernate, Mongoose, Redis, IndexedDB, LocalStorage

Banking and regulatory: Temenos T24 / Transact, T24 enquiries (STANDARD.SELECTION), regulatory reporting (Banque d'Algerie, BEAC / COBAC, BCEAO / UEMOA), prudential reporting, ETL and data lakehouse (Bronze / Silver / Gold), data dictionaries and lineage, electronic document management (ECM / GED), BPM workflow design, Jasper / JRXML document production, Nostro reconciliation

Machine learning and AI: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Vision Transformers (ViT), EfficientNet, ConvNext, TIMM, large language models (LLMs), MediaPipe, OpenCV, TensorBoard

3D graphics and CAD: Three.js, React Three Fiber, OpenCASCADE.js, WebAssembly (WASM), parametric modeling, CAD formats (STEP, IGES, STL, OBJ)

Methodologies: Agile Scrum

CI/CD and tools: Git, Docker, Vite, WebSocket, WebRTC, JWT authentication

Languages: Arabic (native), French (fluent), English (fluent)

PROFESSIONAL EXPERIENCE

Senior Technical Consultant | Tech Lead

ITSS Global | Client missions: Algeria and Congo-Brazzaville | March 2026 - Present

Tech lead on all assigned deliveries below: BNH regulatory reporting and GED, Nostro reconciliation, CEMAC client document production, and CEMAC client T24 regulatory reporting.

- Banque Nationale de l'Habitat (BNH), Algeria - regulatory reporting and electronic document management: led functional cadrage (PV de cadrage), data architecture, and bank-metier enablement so BNH can produce Banque d'Algerie regulatory reports and operate electronic document management on the ITSS Sigma platform (SigmaReports, SigmaGED). Directed the lineage Temenos T24 (JBloader extraction) to Bronze (raw extract) to Silver (SRC_* curated tables and REF_* referentials) to Gold (FACT_* aggregates) to report-canvas restitution, across six domains: Comptabilite, Engagements, Risque, Conformite, COMEX, Exploitation.
- BNH - data dictionaries and traceability: led the master data dictionary and domain-split dictionaries, with template mapping columns (REF, schema column, source table, T24 source, rule / filter). Led reading guides and end-to-end traceability dossiers so non-technical metier can validate canevas cells back to Gold, Silver, Bronze and T24. Incomplete mappings flagged as A COMPLETER; no invented T24 sources or schema columns.
- BNH - reporting packs RRO and M20R: led end-to-end packs from canevas to Bronze, including fictive coherent demo data, SQL pipeline patterns (warehouse prefixes pa_ parameterization, ds_ datasources, cs_ reports), report-platform configuration, and client UI parameterization packs. M20R configuration covers Bilan and Annexes 1 to 7.
- BNH - BPM document workflows: led BPM workshops for the COMEX domiciliation pilot, including process matrices, flux centralisation catalogue, and prerequisite execution tracking (document catalogue, BPM circuits, users, scanners, server). Branching by avis configured at step level; branching by amount or field condition configured as Variantes.
- Nostro reconciliation platform (internally NostroRecon) - delivered and closed: led delivery of the Nostro account reconciliation platform for BNH. Stack: React 19, Vite, Ant Design, Spring Boot 4, Java 21, PostgreSQL 16, T24. Matching profile: MIRRORING (opposite Debit/Credit). Automatic exact-reference matches go VALIDE (SYSTEM); the proposals page holds PROPOSE only. Manual match limit 100 USD. Out of this delivery: internal-account lettrage (phase 2). Project completed.
- CEMAC commercial banking client, Congo-Brazzaville - Temenos Transact document production: led editique delivery, JRXML templates (ED-001 to ED-011) and Document Output configuration (documents, datasets, document types). Live document types include RIB, statements, cash-in and cash-out receipts, repatriation, debit advices, documentary remittance, interest scale, and amortization table. Deployment of Studio compiled Jasper and JRXML templates.
- CEMAC commercial banking client, Congo-Brazzaville - regulatory reporting go-live on Temenos Transact R25 (TAFJ): led T24 enquiries created in the T24 browser (ENQUIRY,INPUT). Delivered TELLER-based enquiries including FX purchases and sales to clients (CG.E.FX.CLIENT.REPORTING), BEAC currency retrocession (LIST.RETROCESSION.BEAC), agency cash position, and BEAC deposits and withdrawals (CG.E.DIS.TXN.AUTH.REPORTING). Designed the NOFILE enquiry path for foreign assets and current DFX needs (ENQ.NOFILE.AVOIRS.EXT.DFX).

Java and Oracle PL/SQL Developer

BFI Groupe | Tunisia | July 2023 - March 2026

- Designed and developed software solutions for financial institutions, in alignment with industry standards and specifications.

- Accounting centralizer for BIAT: integration and synchronization with the bank information systems (core banking and other flows) via the accounting centralizer; orchestration of accounting and transactional flows; integrity and traceability controls; performance optimization (latency, volume).
- Regulatory application for banks in the UEMOA region: design and integration of a prudential reporting solution compliant with BCEAO / UEMOA requirements (collection, normalization, controls, generation and transmission of statements).
- Collaborated with cross-functional teams to analyze requirements, architect scalable solutions, and resolve complex technical issues throughout the development lifecycle.
- Implemented and optimized high-performance Oracle databases, focusing on data integrity and security to support critical transactions (packages, jobs, materialized views, partitioning, indexing, execution-plan tuning).
- Developed custom features including reports, automated workflows, and user access controls tailored to each client's needs.
- Wrote technical documentation including architecture diagrams, API references, and user manuals.

Software Development Engineer

SEGULA Technologies for Groupe Stellantis | France | October 2018 - June 2023

- Led analysis and detailed specification of client requirements to ensure delivery of high-impact solutions.
- Architected and developed internal software for objective calibration of torque converters for automatic transmission, using vehicle dynamics with Excel VBA as the foundation.
- Designed and implemented a benchmarking tool that automatically generated detailed .doc reports according to complex client-defined criteria.
- Engineered an Excel / Access platform for data navigation, visualization, and updating, with external XML import / export and synchronization with an Access database via optimized SQL queries; later migrated to PostgreSQL while keeping backward compatibility with existing tools.

PROJECTS

DermaDetectAI

React, TypeScript, Python, FastAPI, PostgreSQL, PyTorch, TensorFlow, Hugging Face Transformers, Material-UI, Docker | <https://github.com/dhiaBenHmida/DermaDetectAI>

- Full-stack AI diagnostic tool for skin-condition analysis: Vision Transformer (ViT) feature extraction and custom-trained MLP classifier.
- MedGemma conversational diagnosis, role-based access control (Admin, Doctor, Super Admin), GPU-accelerated inference, patient and session dashboards, PWA support.
- Separate training repository: <https://github.com/dhiaBenHmida/DermaDetectAI-Training>

CADSHIP - 3D Model Marketplace

React, Three.js, React Three Fiber, Node.js, Express, MongoDB, OpenCASCADE.js, WebAssembly, JWT, Redis | <https://github.com/dhiaBenHmida/CADSHIP>

- Full-stack marketplace for sharing, buying, and selling 3D CAD models, with visualization for STEP, STL, OBJ, FBX, GLTF, and GLB.
- Real-time collaboration (WebRTC, Socket.io), e-commerce, JWT authentication, GridFS storage, Redis caching.

VisionForge AI - Centralized Computer Vision Framework

React, TypeScript, Spring Boot, Python, FastAPI, PyTorch, Vision Transformers, EfficientNet, ConvNext, MediaPipe, OpenCV, Docker | <https://github.com/dhiaBenHmida/VisionForge-AI>

- Extensible computer-vision training and deployment platform with a unified inference interface (adapters), model registry, versioning, rollback, and A/B testing.
- MediaPipe quality validation, multi-model training pipeline, hybrid Spring Boot and FastAPI microservices.

SkinSmart - Skincare Recommendation System

React, TypeScript, Material-UI, TensorFlow.js, PHP, MySQL | <https://github.com/dhiaBenHmida/SkinSmart>

- Interactive survey with conditional branching, webcam skin analysis (TensorFlow.js), personalized product recommendations, and e-commerce (cart, orders, PHPMailer).
- Admin dashboard for products, diagnoses, questions, orders, and A/B testing.

EDUCATION

National Engineering Degree

École Nationale Supérieure d'Ingénieurs de Tunis (ENSIT) | Tunis, Tunisia | September 2015 - June 2018

Preparatory Classes for Engineering Studies (MP track)

Institut préparatoire aux études d'ingénieurs d'El Manar (IPEIEM) | Tunis, Tunisia | 2013 - 2015

Tunisian Baccalaureate - Mathematics specialty

2013

TRAINING AND CERTIFICATIONS

Accelerated VBA Training - Stellantis

Technical Center, Carrières-sous-Poissy | August 2019

Intensive hands-on training on advanced VBA development in industrial applications.

Massinissa Framework Training - BFI

Duration: 4 months

Four-month program on the Massinissa framework architecture, based on Java and MVC.